

PIV Project

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1 Motivations

This project aims to estimate geometric transformations between a reference image and a sequence of images using classical and modern computer vision techniques. The project is structured in two parts:

- *Part 1* : Estimation of 2D homography from grayscale images
- *Part 2* : Estimation of rigid 3D transformation (rotation and translation) from RGB-D data

These techniques find applications in object tracking, 3D reconstruction, robotic localization, and augmented reality.

2 Part 1: Planar Object Tracking via Homography

This section describes the implementation of a 2D planar tracking system that estimates the Homography (H) between a reference template and a sequence of video frames.

2.1 Algorithmic Details

2.1.1 Feature Extraction and Matching

The system utilizes the **Scale-Invariant Feature Transform (SIFT)** to :

- Detect interest points robust to scale and rotation
- Extract 128-dimensional invariant descriptors
- Generate a '.mat' file per frame containing keypoints and descriptors

SIFT Parameters : OpenCV defaults ($\sigma = 1.6$, standard octaves and scales)

Then, feature matching is performed between the template and each frame using the following strategy:

We use Euclidean distance metric combined with **Lowe's ratio test** to filter ambiguous matches: "" For each descriptor in the frame: - Find the 2 nearest neighbors in the template - If $\text{distance}(\text{NN1}) < 0.75 \times \text{distance}(\text{NN2})$, accept the match "" The threshold of 0.75 is chosen based on Lowe's original paper to balance precision and recall.

2.1.2 Homography Estimation (DLT with Normalization)

The relationship between planar points in the template ($\mathbf{x}$) and the frame ($\mathbf{x}'$) is modeled by a 3×3 Homography matrix H such that $\mathbf{x}' \sim H\mathbf{x}$. The **Direct Linear Transform (DLT)** algorithm is used to solve for H using Singular Value Decomposition (SVD).

Data Normalization A crucial step implemented is isotropic normalization. Before applying DLT, points are translated so their centroid is at the origin and scaled so the average distance from the origin is $\sqrt{2}$. This improves the numerical stability of the SVD.

$$T_{norm} = \begin{bmatrix} s & 0 & -s \cdot c_x \\ 0 & s & -s \cdot c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Matrix A Construction For N correspondence points $(x_i, y_i) \rightarrow (u_i, v_i)$, create a $2N \times 9$ matrix:

$$A_i = \begin{pmatrix} -x_i & -y_i & -1 & 0 & 0 & 0 & x_i u_i & y_i u_i & u_i \\ 0 & 0 & 0 & -x_i & -y_i & -1 & x_i v_i & y_i v_i & v_i \end{pmatrix}$$

SVD and H Extraction $H = VT^{-1}$, where V comes from complete SVD

Denormalization The final Homography is denormalized via:

$$H = T_{dst}^{-1} H_{norm} T_{src} \quad (2)$$

2.1.3 Robust Estimation (RANSAC)

To handle mismatches effectively, a Random Sample Consensus (RANSAC) scheme is implemented with the following parameters:

- **Iterations:** 5000.
- **Sample Size:** 4 points (minimum required for DLT).
- **Inlier Threshold:** 3.0 pixels.
- **Refinement:** After identifying the best model, H is re-computed using **all** inliers found by the best model to minimize the reprojection error via least squares.

2.2 Experimental Setup and Evaluation

Dataset We used the TAAG dataset, which provides a reference template image and a sequence of frames with varying perspectives and occlusions.

Evaluation metrics We considered the *number of inliers* as an indicator of matching quality and *robustness* as an ability to handle partial occlusions and significant transformations

Execution Pipeline For every frame:

1. Features are extracted and matched against the pre-computed template features.
2. RANSAC estimates the best H .
3. The result is stored as a `.mat` file containing the 3×3 matrix.

2.3 Example Results

The algorithm successfully tracks the template even under perspective distortion. The visualization (conceptually) involves projecting the four corners of the template onto the current frame using H .

2.3 Example Results

****Typical result for a frame:**** - ****Raw Matches (after ratio test):**** 50-100 correspondences - ****RANSAC Inliers:**** 40-80 (80-85 acceptance rate) - ****Average Reprojection Error (inliers):**** < 1.5 pixels

Output File: `homography_0001.mat` contains the H matrix (3×3) representing the projective transformation

****Observations:**** - Point normalization is crucial for numerical stability - RANSAC is robust to residual false matches - Estimated homographies enable template reprojection on each frame with sub-pixel accuracy

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3 Part 2: 3D Rigid Body Motion Estimation

This section details the estimation of the camera's 3D trajectory (Odometry) by calculating the Rigid Body Transformation (R, T) between consecutive frames.

3.1 Algorithmic Details

3.1.1 Feature Extraction and Matching

- Use SIFT on RGB images to identify 2D-2D correspondences
- Apply same Lowe's ratio test (threshold = 0.75)
- Recover SIFT descriptors robust to lighting changes

3.1.2 3D Point Generation

Unlike Part 1, this module utilizes RGB-D data. 2D keypoints (u, v) are back-projected into 3D space (X, Y, Z) using the corresponding depth map and the Pinhole Camera Model intrinsics matrix K :

$$Z = \text{depth}(v, u) \quad (3)$$

$$X = \frac{(u - c_x) \cdot Z}{f_x}, \quad Y = \frac{(v - c_y) \cdot Z}{f_y} \quad (4)$$

Points with invalid depth ($Z \leq 0$) are discarded.

3.1.3 Orthogonal Procrustes Analysis

To find the rotation R and translation T that align two clouds of 3D points ($P_B \approx R \cdot P_A + T$), the Procrustes method is used.

- **Centering:** Centroids of both point clouds are subtracted.
- **SVD:** The cross-covariance matrix $H = P_B^T P_A$ is decomposed into U, Σ, V^T .
- **Rotation:** $R = UV^T$.
- **Reflection Correction:** To ensure a valid rotation (determinant $+1$) rather than a reflection (determinant -1), if $\det(R) < 0$, the last row of V^T is negated.
- **Translation:** $T = \bar{P}_B - R \cdot \bar{P}_A$, where $\bar{P}_A$ and $\bar{P}_B$ are the centroids of the original point clouds.

3.1.4 RANSAC for 3D Alignment

A RANSAC loop wraps the Procrustes method to be robust against depth noise and matching errors:

- **Iterations:** 1000.
- **Sample Size:** 3 points (minimum for rigid alignment).
- **Threshold:** 0.01 (meters), applied to the Euclidean distance between transformed source points and target points.

3.2 Experimental Setup and Evaluation

Dataset

- 1 RGB template + depth (`template_depth.mat`)
- 12 RGB frames + depth (`frame_NNNN.mat`)
- Resolution: typically 640×480 pixels
- Depth format: $H \times W$ matrix with values in millimeters

Evaluation metrics

- **Number of Valid 3D Correspondences:** Indicator of matching quality and depth density
- **RANSAC Inliers:** Ratio of robustness in pose estimation
- **Average 3D Error:** For inlier points

Execution Pipeline The system iterates through pairs of `frame_RGB` and `frame_Depth`.

1. Load RGB template + depth
2. For each frame:
 - Extract and match SIFT features
 - Back-project matched 2D points to 3D using depth maps
 - Run RANSAC + Procrustes to estimate R, T
 - Store results in `.mat` files containing R and T

3.3 Example Results

The resulting R and T matrices describe the camera's motion. By concatenating these transformations, the global trajectory of the camera can be plotted. The use of RANSAC significantly reduces "jumps" in the trajectory caused by noisy depth values at object boundaries.

3.3 Example Results
Typical result for a frame: - **2D-2D Correspondences:** 60-150 SIFT matches - **Valid 3D Correspondences** (depth available): 50-120 - **RANSAC-Procrustes Inliers:** 40-100 (70-85 acceptance) - **Average Error (inliers):** 1-3 cm

Output File: `transform_NNNN.mat` contains R (3×3 orthogonal rotation matrix) and T (3×1 translation vector)

****Validation:**** - Matrix R satisfies $R^T R = I$ (orthogonality) - $\det(R) = 1$ (pure rotation, no reflection) - Vector T represents translation in meters (or depth units)

4 Discussion and Limitations

4.1 Comparison of Methods

- **2D Homography (Part 1):** Efficient for tracking planar objects. However, it fails if the target object is non-planar or if the camera motion induces significant parallax that a planar model cannot represent. The DLT normalization implementation is critical for maintaining accuracy when coordinates are large.
- **3D Procrustes (Part 2):** Provides full 6-DoF pose estimation. It requires depth data, making it dependent on the quality of the depth sensor. The implementation correctly handles coordinate system reflections, which is a common pitfall in SVD-based estimation.

4.2 Identified Limitations

Part 1

- Limited to planar scenes
- Sensitive to lighting changes (though SIFT mitigates this)
- Requires sufficient extractable features

Part 2

- Critical dependency on depth data quality (noise, holes)
- Degraded performance if valid 3D correspondences < 3 (RANSAC MSS)
- Sensitivity to calibration of K

4.3 Conclusion

The implemented pipeline demonstrates a standard computer vision approach: Feature Extraction $\rightarrow$ Matching $\rightarrow$ Robust Model Estimation. The use of SVD for both DLT and Procrustes ensures mathematically rigorous solutions to the minimization problems.